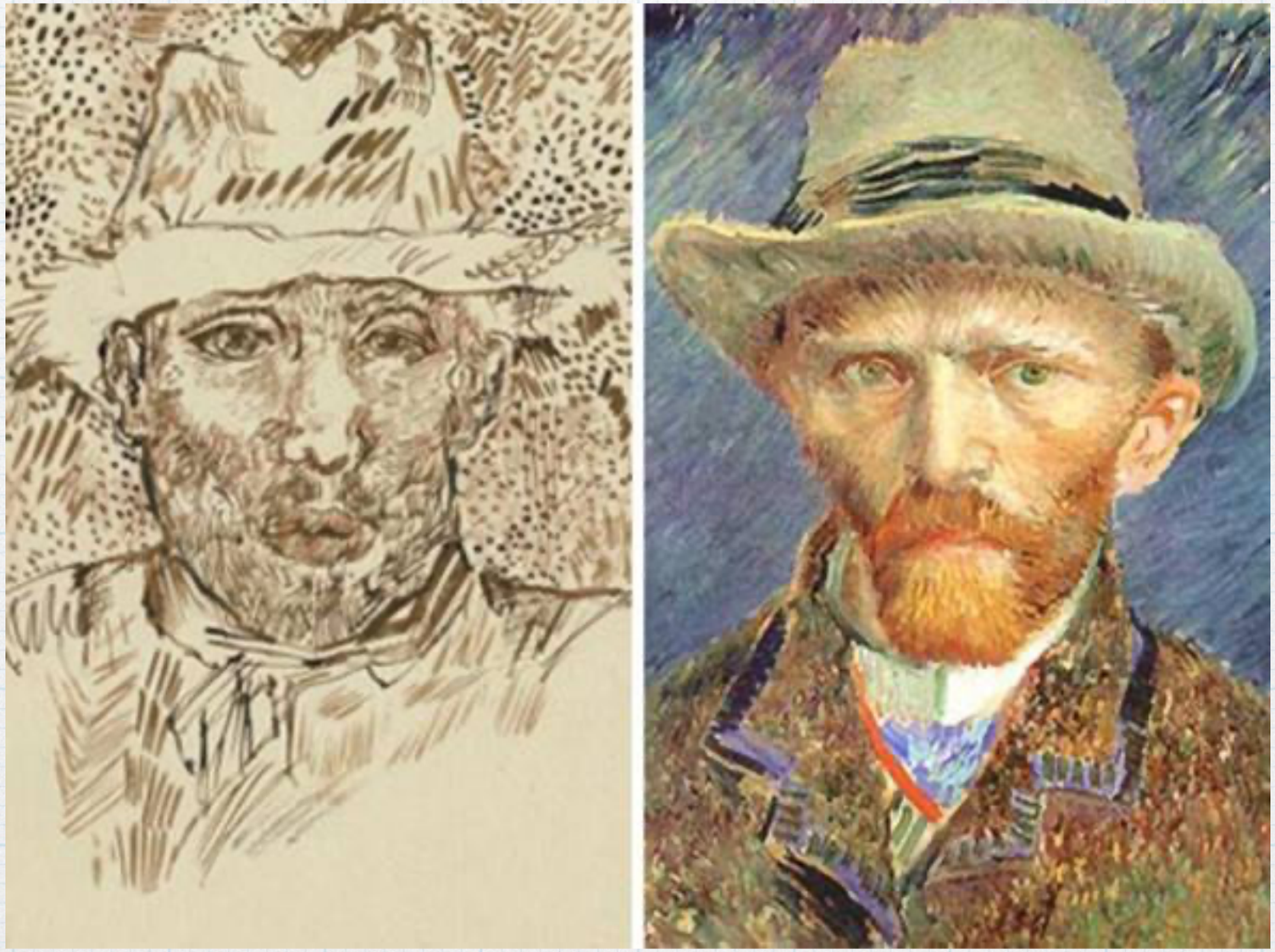
Modeling optimization problems

Making a portrait



Challenges of this module

* How do we represent an optimization problem?
* What are the "tricks" to represent decisions?
* What are the different types of constraints?
* How can we transform the the problem?
* How to deal with problems with multiple optimization criteria?
* What about no optimization criteria?
* ...

One recommendation

We are only modeling

Refrain from thinking about solving the problem

even if it is obvious

Modeling an optimization problem

* **Decisions**
  What are the decisions to be taken?
  How do you characterize your solution?
* **Rules**
  How decision are regulated?
* **Objective**
  What do you have to optimize?

Modeling an optimization problem

* Decisions
  Variables
* Rules
  Mathematical constraints
  equalities or inequalities
* Objective
  Function

Variables: links with previous knowledge

* variable: unknown value
  it has to be decided

* variables -> decisions

Variables

* Non negative value variables
* Unrestricted value variables
* Relative value variables
* 0-1 variables
* Discrete value variables

Section 5.1

Variables: non negative absolute quantities

x_A = how many phones of type A must be produced

x_B = how many phones of type B must be produced

Negative values are not allowed

Variables: unrestricted absolute quantities

Typical applications

Manage settings where quantities are unrestricted in sign (e.g. temperatures)

Manage perturbations to a current solution

Example: timetabling

Produce the **train timetable** for the next year

Given

Set of trains $T = \{1, \ldots, n\}$

Set of stations $S = \{1, \ldots, m\}$

Old timetable: arrival times a_{ts} $\forall t \in T, s \in S$

Perturb current arrival times

π_{ts}

Unrestricted variables

How to transform an unrestricted variable into a restricted one

$$\pi^{+}_{ts} \geq 0$$

$$\pi^{-}_{ts} \geq 0$$

$$\pi_{ts} = \pi^{+}_{ts} - \pi^{-}_{ts}$$

Variables: relative values

Example: capital investment problem

Given a budget: 137k€

Given a set of financial instruments

	1	2	3	4	5
max amount	40	12	130	5	400
coupon	3.2	1.5	4.2	0.7	17

Decide how much to invest in each instrument

Maximize the economical return from the coupons

Variables: relative values

Decisions

the fraction of each instrument to be bought

$$0 \leq y_i \leq 1, \qquad i = 1, \ldots, n$$

Try to find the optimal solution

Groups of 3-4 persons

Elaborate a strategy

5 minutes

Formative assessment

Kahoot!

Solution strategy

$$\max \sum_{i=1}^{n} c_i y_i$$

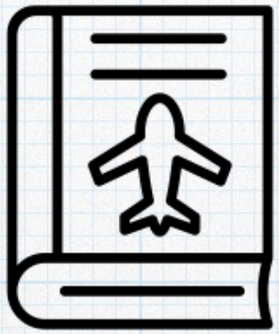
$$\sum_{i=1}^{n} a_i y_i \leq b$$

$$0 \leq y_i \leq 1 \quad i = 1, \ldots, n$$

sort indices so that $i<j \Rightarrow c_i / a_i \geq c_j / a_j$

```
i←1;
repeat
        y_i ← min {1, b/a_i};
        b←b-y_i a_i;
        i←i+1;
until b=0 or i>n
```

Self reflection



prove that the strategy described before always produces the optimal solution for any n, c, a, b

Logical variables (0-1)

Restrict a **relative value variable** to assume value **either 0 or 1**

$$x_i \in \{0, 1\}$$

$x_i = 0$ $\Rightarrow$ do not take item i

$x_i = 1$ $\Rightarrow$ take item i

Example

A burglar sneaks into a jewelry

The burglar has a bag having capacity of **8 kg**

Objects

		1	2	3	4	5	6
value	v_i	7	2	4	5	4	1
weight	w_i	5	3	2	3	1	1

Select the objects to be stolen

Maximize the value

Do not overload the bag

Adapt the strategy seen for the continuous case

does it aways find the optimal solution?

Solution strategy

$$\max \sum_{i=1}^{n} c_i y_i$$

$$\sum_{i=1}^{n} a_i y_i \leq b$$

$$\cancel{0 \leq y_i \leq 1} \quad i = 1, \ldots, n \qquad y_i \in \{0,1\}$$

sort indices so that $i<j \Rightarrow c_i / a_i \geq c_j / a_j$

```
i←1;
repeat if b≥a_i then y_i←1 else y_i←0
        ~~y_i ← min{1, b/a_i}~~;
        b←b - a_i y_i;
        i←i+1;
until b=0 or i>n
```

Compare the two problems

P1:
$$\max \sum_{i=1}^{n} c_i y_i$$
$$\sum_{i=1}^{n} a_i y_i \leq b$$
$$0 \leq y_i \leq 1 \quad i = 1,\ldots,n$$

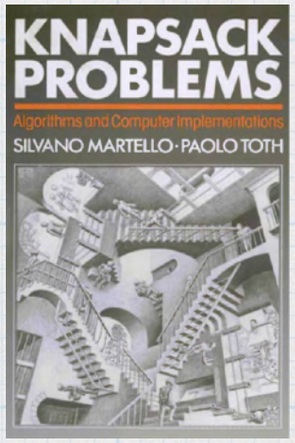
"easy"

P2:
$$\max \sum_{i=1}^{n} c_i y_i$$
$$\sum_{i=1}^{n} a_i y_i \leq b$$
$$y_i \in \{0,1\} \quad i = 1,\ldots,n$$

"difficult"

Write your self reflections

- What is the order of magnitude of the **number of feasible solutions** of P1 and of P2?
- What does it make P1 "easier" to solve with respect to P2?



http://www.or.deis.unibo.it/kp/KnapsackProblems.pdf

Logical variables

Complement

$$x_i' = 1 - x_i$$

Discrete value variables

Integer variables

e.g. in the cell phone assembly problem we could restrict variables to assume integer values

$$x_A, x_B \in \mathbf{Z}_+$$

Binary variables are a particular case of integer variables

Discrete value variables

In many problems we select the value of a **variable** within a **finite set**

e.g. the memory size of a computer, the capacity of a telecommunication link

$$x \in \{v_1, v_2, \ldots, v_r\}$$

Constraints

* Availability constraints
* Requirement constraints
* How to transform constraints
* Blending constraints
* Logical constraints (0-1 and continuous case)
* Flow conservation constraints

Section 5.2

Rules = Constraints

Resource availability constraints

The number of used components cannot exceed the available ones

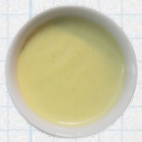
$1x_A + 2x_B \leq 10$	display
$2x_A + 2x_B \leq 18$	memory
$1x_A + 3x_B \leq 12$	transmission
$2x_A + 3x_B \leq 21$	keyboard
$1x_A \leq 9$	navigation
$1x_A \leq 10$	camera
$x_A, x_B \geq 0$	non negativity

Constraints

Requirement constraints

Example: poke bowl diet

Set of food types

i									
c_i	300	250	35	40	180	100	200	150	400

Minimum required calories: 700

Combine the foods so as 700 calories are reached at least

Transformations

$$\sum_{i=1}^{n} a_i x_i \leq b$$

$$\Downarrow$$

$$\sum_{i=1}^{n} -a_i x_i \geq -b$$

What about equality?

Blending constraints

Typical in oil or steel industry

requirements on the percent presence of certain components

e.g. diet: share coming from fruits and vegetables ≥30%

Logical constraints

binary variables can be seen as logical variables

they can be used to express **propositional logic** operators

$\vee \quad \rightarrow \quad +$

$\wedge \quad \rightarrow \quad \cap$

SAT

Given a logical formula
Find a truth assignment that makes it true

Conjunctive Normal form

$$(\pm P_{11} \vee \ldots \vee \pm P_{1r}) \wedge (\pm P_{21} \vee \ldots \vee \pm P_{1q}) \wedge \ldots \wedge (\pm P_{m1} \vee \ldots \vee \pm P_{ms})$$

literal

clause

at least one literal must be true for each clause

SAT

Literal $\longleftrightarrow$ binary variable
direct or complemented

P $\qquad$ x

$\neg P$ $\qquad$ $1 - x$

Clause $\longleftrightarrow$ inequality

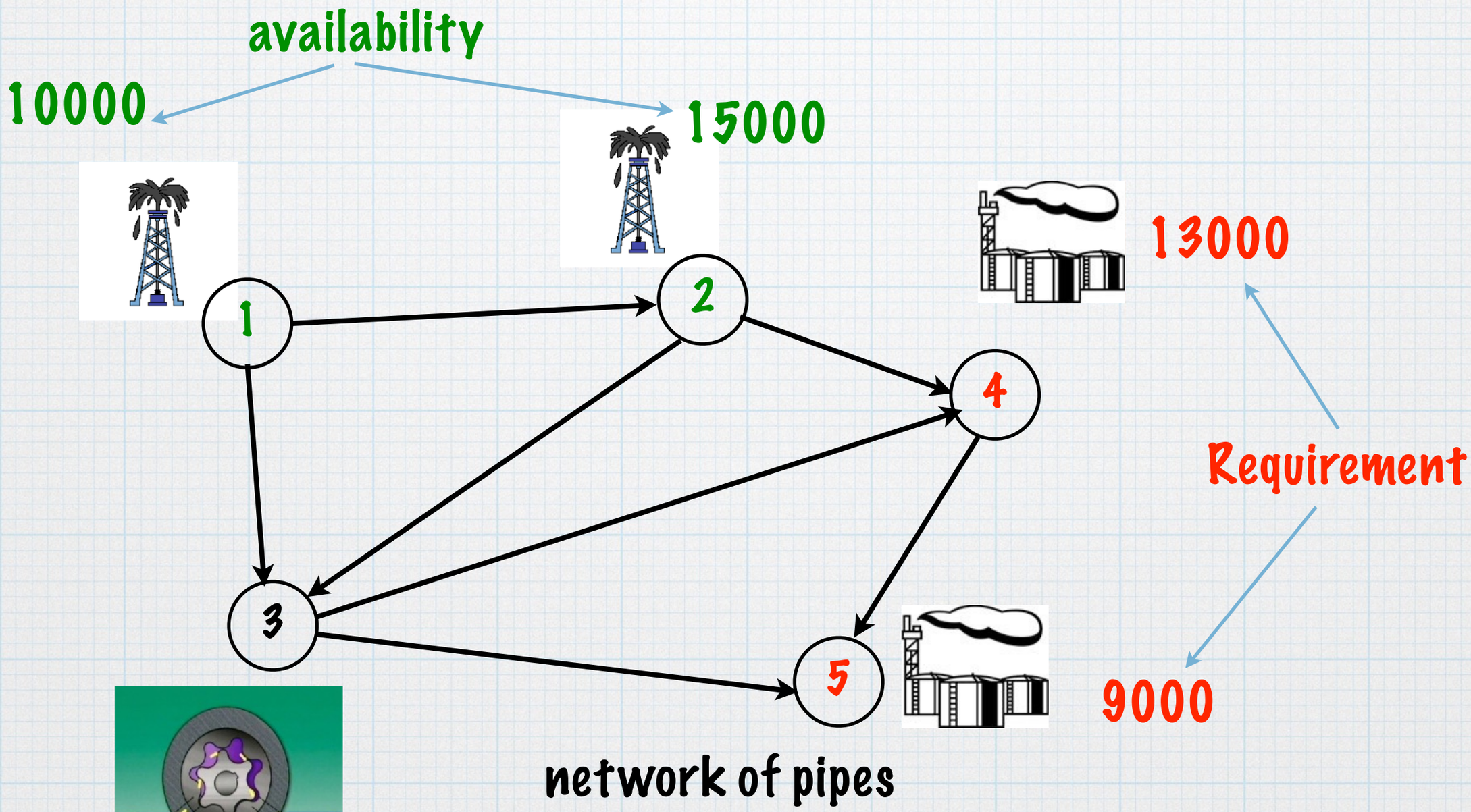
Section 5.2: Satisfiability*

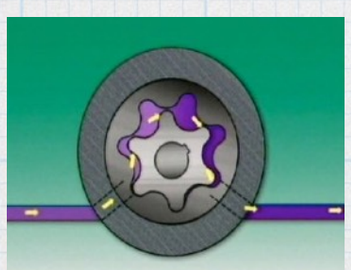
Extend logical constraints to real variables

$$x_{salmon} = 0 \quad \textbf{or} \quad x_{chicken} = 0$$

Flow constraints

Example: oil field





Distribute the oil (flow) along the pipes

Objective function

* minimum
* maximum
* max-min, min-max
* no objective function
* multiple objective functions

Section 5.3

Objective function

Maximize the overall score

$$\max 3x_A + 8x_B$$

Minimize the cost

$$\min \sum_{(i,j)\in P} c_{ij}x_{ij}$$

Transform a min into max and vice-versa

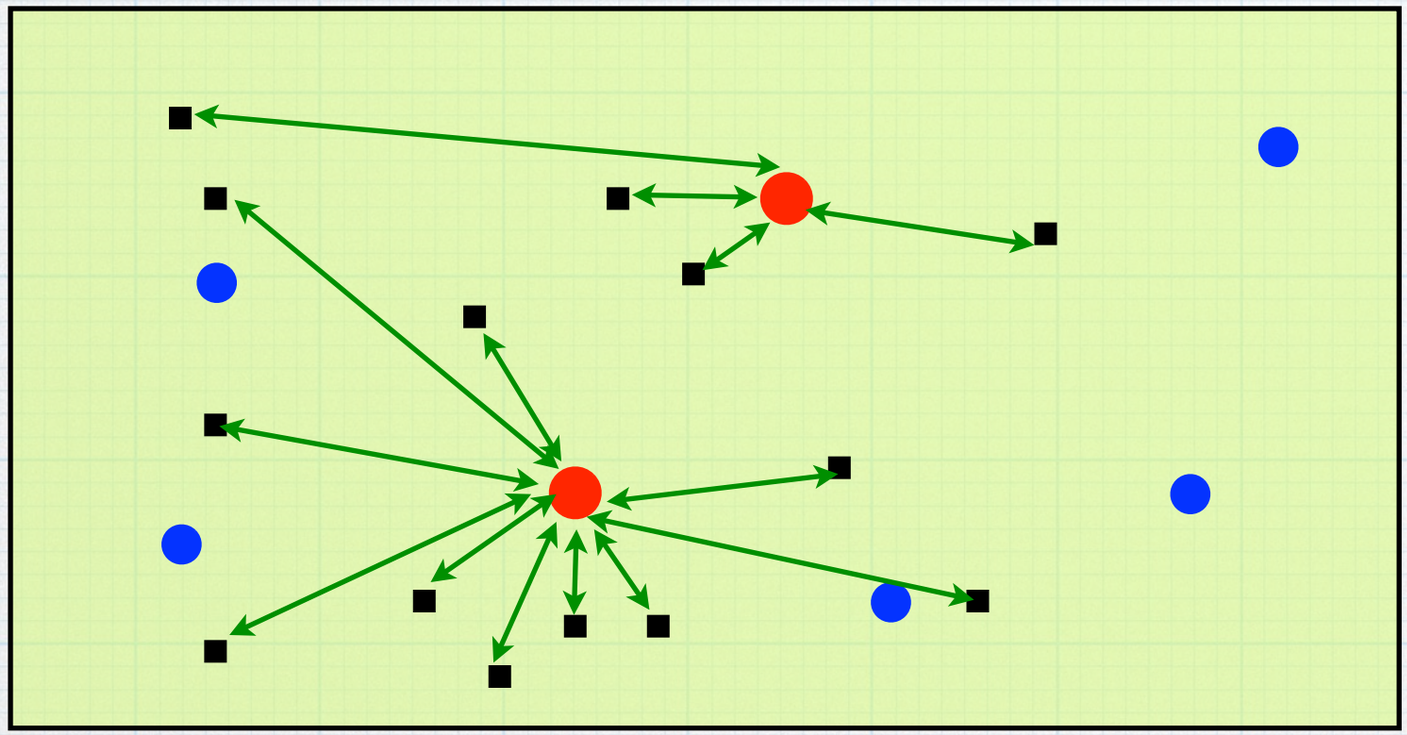
Objective function

Problems without an objective function

Use a constraint as an objective

Objective function: bottleneck

Base Station location problem



Candidate site

Customer

Which base stations to activate

Assign customers to activated base stations

For each connection customer-base station we know the emission power

$$p_{ij}$$

Capacity of base stations (max number of customers) M

Minimize the maximum emission power

Multiple objective functions

Base station location problem

Full coverage is no more required

An installation cost for each base station is considered

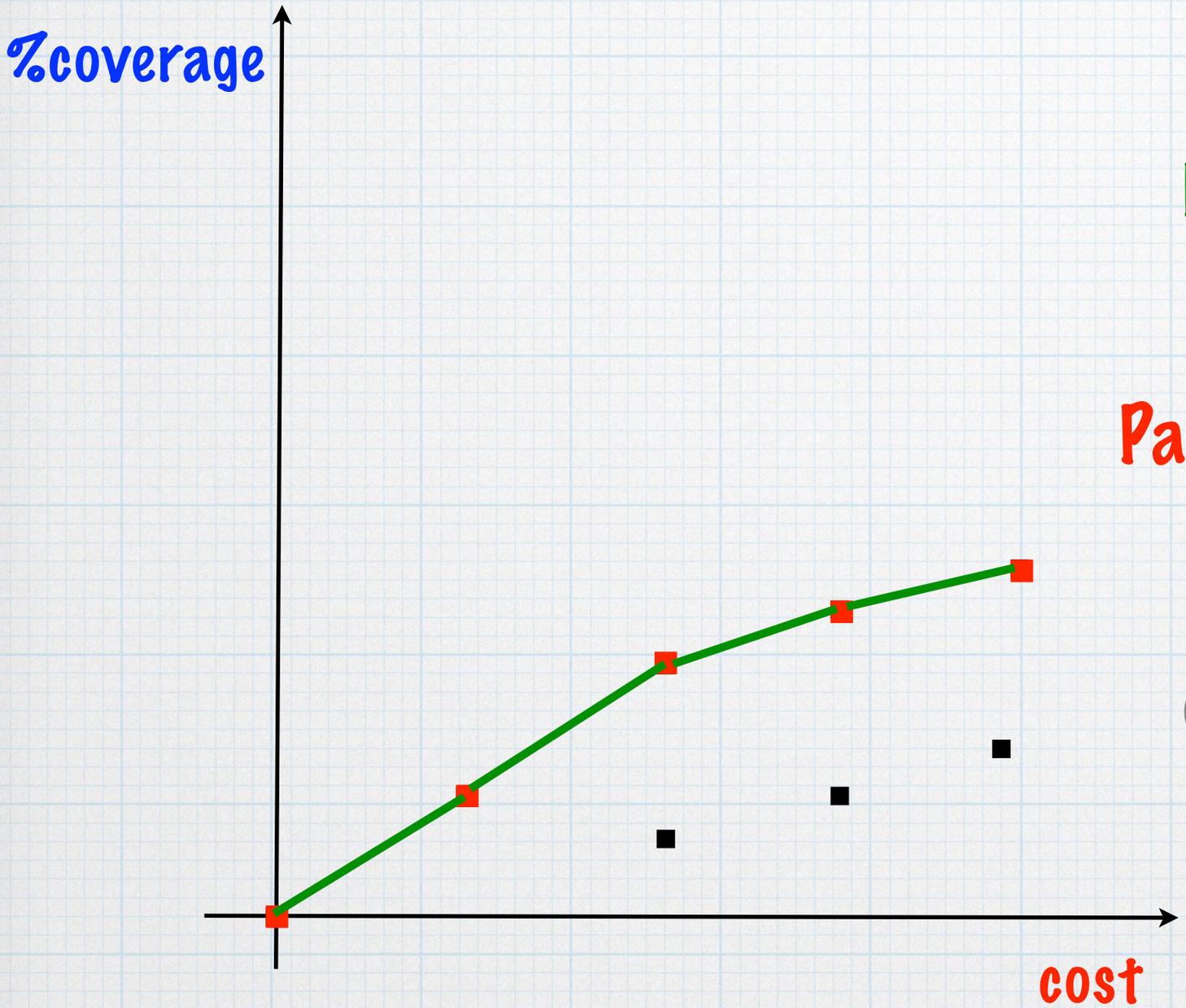
c_j

two conflicting objective functions

Maximize the coverage

Minimize the installation cost

Multicriteria analysis



Efficiency frontier

Pareto optimal solutions

dominated solutions

Optimization in cancer radiation treatment

BEEP

ILO of this module

* variable types
* constraint types
* objective function types
* equivalence transformations (variables, constraints and objective functions)

Chapter 1